**School or Department:** SEG

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**Module Code:** E211

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**Assessment:** ESE

Assessment Document:

Assessment Paper AY 2024 Semester 2 E211 ESE ...

Section A (12 Questions)

**Q1.** At a specialist clinic, patients arrive according to fixed appointment times. In this queuing system, the population source would be _____ and the arrival pattern would be _____.

- fixed, random
- infinite, non-random
- finite, non-random (Correct Option)
- finite, Poisson

**Q2.** Which of the following methods is used to solve a Maximum Flow problem?

- Ford Fulkerson's algorithm (Correct Option)
- Clarke and Wright savings method
- Distribution model
- Additive Decomposition method

**Q3.** Referring to Figure 1a, what type of correlation can be seen between the 2 variables in the chart?

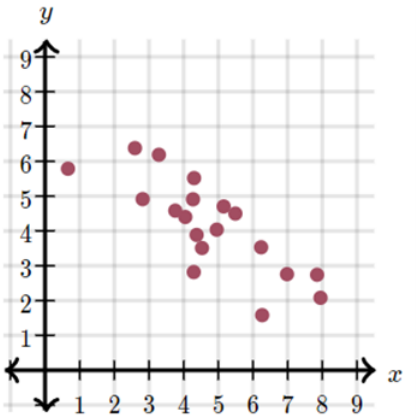


Figure 1a: A Time Series Chart

- Positive correlation
- Negative correlation (Correct Option)
- Random correlation
- No correlation

**Q4.** Find a routing solution by applying the Nearest Neighbour Method to the network in Table 1. **Assume starting node is B.**

Table 1: Travel Distance between Nodes

| | A | B | C | D |
|---|---|---|---|---|
| **A** | | 8 | 5 | 13 |
| **B** | 8 | | 11 | 9 |
| **C** | 5 | 11 | | 14 |
| **D** | 13 | 9 | 14 | |

- A – C – B – D
- B – A – C – D – B (Correct Option)
- B – C – B – D – A
- B – D – A – C – B

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**Q5.** Which one of the following is **NOT** a pitfall when conducting a simulation study?

- Failing to define the objectives and scope of the simulation study clearly
- Expecting the simulation model to automatically provide an optimal solution to the problem
- Neglecting to validate the simulation model before running experiments
- Gathering detailed data on the system through time studies and interviews (Correct Option)

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**Q6.** Referring to the statistics obtained from FlexSim simulation (Figure 1b), how many items have passed through Queue1?

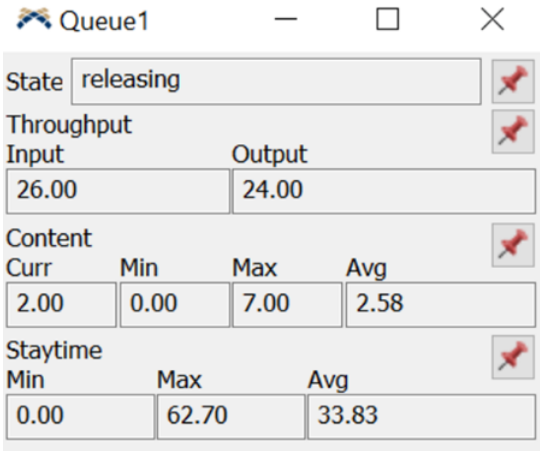


Figure 1b: Queue1 Statistics from FlexSim

- 7
- 2
- 26
- 24 (Correct Option)

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**Q7.** Referring to the menu bar (Figure 1c) in FlexSim software, what will happen if you uncheck 'Repeat Random Streams' under Statistics?

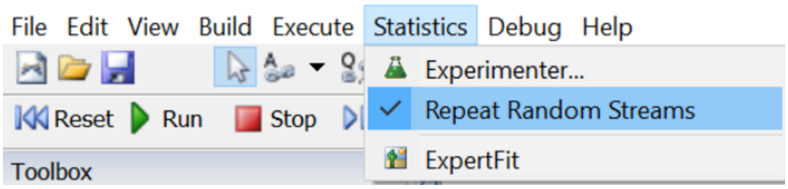


Figure 1c: FlexSim Menu

- Results obtained will be different in each simulation run. (Correct Option)
- There will be no randomness in the simulation.
- It will affect Experimenter results.
- Simulation results will become more consistent.

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**Q8.** *Show your workings clearly*.

An apparel manufacturer ships products from its warehouse to 6 other locations in the country. The distribution network for its products is shown in Figure 2a. The nodes represent locations and values on the arcs represent distances between locations.

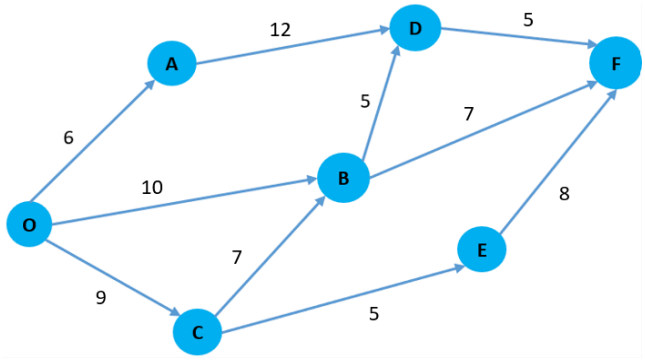


Figure 2a: A Distribution Network

**Q8a.** Table 2a shows the partial workings to find the shortest path from the warehouse (node O) to each of the other locations (Nodes A to F).

Table 2a: Partial Workings for Shortest Path Problem

| Iteration | Solution Set | Arcs | Total Distance |
|---|---|---|---|
| Iteration 1 | O | O → A<br>O → B<br>O → C | 6<br>10<br>9 |
| Iteration 2 | O<br>A (6) | O → B<br>O → C<br>A → D | 10<br>9<br>18 |
| Iteration 3 | O<br>A (6)<br>C (9) | O → B<br>A → D<br>C → B<br>C → E | 10<br>18<br>**V** 16<br>14 |
| Iteration 4 | O<br>A (6)<br>C (9)<br>B (10) | A → D<br>C → E<br>B → D<br>**W1** | 18<br>14<br>15<br>**W2** 17 |
| Iteration 5 | … | … | … |

**Q8a(i).** Calculate the value of **V** in Table 2a.

V= 9+7 =16

**Q8a(ii).** In Iteration 4, what is the arc represented by **W1**? State the corresponding Total Distance represented by **W2**.

B->F. Total Distance (W2) = 17

**Q8a(iii).** Which node should be added into the solution set in **Iteration 5**? Explain.

Node E as it has the next lowest total distance of 14.

**Q8a(iv).** State the shortest path from node O to node D <u>and</u> the total distance of the shortest path from O to D.

Shortest path: O -> B -> D. Total distance = 10 + 5 = 15

**Q8b.** Figure 2b shows the distribution network of a product. The value at each arc represents the **shipping capacity** along each route.

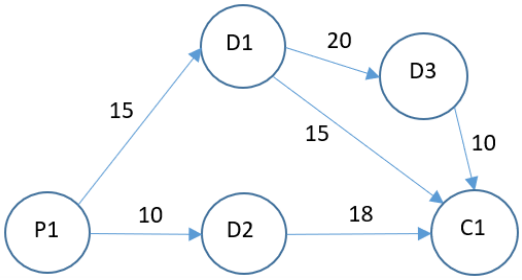


Figure 2b: Product Distribution Network

By following the below paths sequentially, state the flow quantity through each path and determine the maximum flow quantity that can be shipped

from P1 to C1.

P1 -> D1 -> D3 -> C1: 10

P1 -> D2 -> C1: 10

P1 -> D1 -> C1: 5

Total maximum flow quantity = 25

**Q8c.** A distributor ships a new product from a warehouse at Node A. Customers are located at Nodes D & E. The distribution network for shipping the product is shown in Figure 2c where the values on the arcs represent unit shipping costs. The manufacturer wants to determine the lowest cost shipping plan by formulating a Linear Programming (LP) model to solve the Minimum Cost Flow problem.

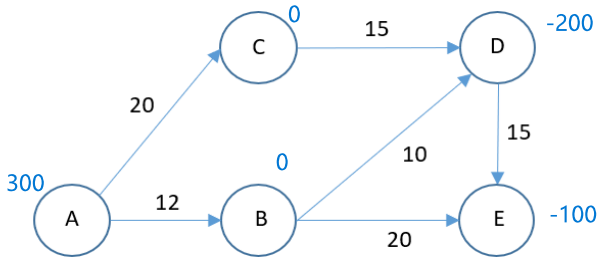


Figure 2c: Distribution Network

The decision variable Xij is defined as the quantity of products to ship from node i to node j (i, j = A, B, C, D or E).

Given:

- Node A can supply up to 300 units
- Nodes D and E have demands of 200 and 100 units respectively
- Nodes B and C are transhipment hubs with zero supply and demand

**Q8c(i).** How many decision variables will the LP model contain for the Minimum Cost Flow model in Figure 2c?

6; six

**Q8c(ii).** Using the decision variables defined above, formulate the following constraints:1.

Net flow constraint for **Node D**.

2. Maximum shipping capacity from Node B to E is 80.

3. Net flow constraint for **Node A if the demand at Node D is reduced from 200 to 150**.

1. Node D: XDE - XCD - XBD = -200
2. XBE <= 80
3. XAB + XAC <= 300

**Q8d.** A product is to be delivered from Depot (A) to a number of stores (B, C, D and E). A truck will leave the Depot (A), visit each store once and return to the Depot after delivery. Travel distances between locations are given in Table 2b.

Table 2b: Travel Distances Between Locations

| From | To (Travel distance in km) | | | | |
|---|---|---|---|---|---|
| | A | B | C | D | E |
| A | - | 18 | 9 | 23 | 15 |
| B | 18 | - | 12 | 17 | 14 |
| C | 9 | 12 | - | 10 | 22 |
| D | 23 | 17 | 10 | - | 11 |
| E | 15 | 14 | 22 | 11 | - |

**Clark and Wright savings method** is used to plan the delivery route. Table 2c shows the **Savings** in distance (km) between 2 stores ranked in descending order.

Table 2c: Savings Ranked in Descending Order

| Rank | Savings | From | To |
|---|---|---|---|
| 1 | 27 | D | E |
| 2 | 24 | B | D |
| 3 | 22 | C | D |
| 4 | 19 | B | E |
| 5 | 16 | C | E |
| 6 | ? | B | C |

A-D-E-A
A-B-D-E-A
X
X
A-B-D-E-C-A
X

**Q8d(i).** Calculate the savings between nodes B and C.

Savings (BC) = AB + AC - BC = 18 + 9 - 12 = 15

**Q8d(ii).** Referring to Table 2c, at Rank 1, a sub-tour of A-D-E-A is constructed. What step should be carried out next at Rank 2?

Combined B-D into A-D-E-A since D is adjacent to A. So it will be A-B-D-E-A.

**Q8d(iii).** State the complete route based on Clark and Wrights Savings Method and calculate the total travel distance of the route.

A-B-D-E-C-A. Total distance = 18 + 17 + 11 + 22 +9 = 77km

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**Q9.** *Show your workings clearly and round off your answer to* ***2 decimal places*** ***where necessary.***

Katie works in a company that sells health and wellness products. Her role in the company is sales and demand planning.

**Q9a.** Sales of an existing air purifier has been stagnating recently. To predict customer demand for the next period, Katie collected past sales data (Table 3a) to carry out a forecast.

Table 3a: Historical Sales from Past 6 Months

| Month | Sales Figure (Units) |
|---|---|
| 1 | 80 |
| 2 | 85 |
| 3 | 72 |
| 4 | 77 |
| 5 | 69 |
| 6 | 79 |

**Q9a(i).** Calculate the forecast for Month 7 sales using **3-month** moving average.

(77+69+79)/3 = 75

**Q9a(ii).** Calculate the forecast for Month 7 sales using Exponential Smoothing method. Assume smoothing constant is 0.4 and forecast for Month 6 sales is 74.16.

Forecast (month 7) = (0.4*79) + (1-0.4)*(74.16) = 76.10

**Q9a(iii).** Which of the above forecast methods is more responsive to recent changes in the sales data? Explain briefly.

**Q9a(iv).** A colleague suggested that Katie should use Holt's method for the above data forecasting. Do you agree? Explain why.

**Q9a(v).** The company has developed a prototype of a new air purifier. Explain how grassroots forecasting method can help predict the demand for the new product.

**Q9b.** Katie developed an Excel template for Multiplicative Decomposition Method (Table 3b) to help her forecast the demand for a seasonal product.

Table 3b: Workings of Multiplicative Decomposition Method

| Month/ Week | Actual Value (Y) | Time Period (X) | Moving Average | Centred Moving Average | Seasonal Ratio | Seasonal Index | Deseasonalized value | Deseasonalized forecast | Seasonalized forecast |
|---|---|---|---|---|---|---|---|---|---|
| 1/1 | 742 | 1 | | | | 1.627 | 455.915 | 462.396 | 752.547 |
| 1/2 | 467 | 2 | 521.75 | | | 0.880 | 530.959 | 489.064 | 430.151 |
| 1/3 | 378 | 3 | 568 | 544.875 | 0.694 | 0.701 | 539.496 | 515.732 | 361.349 |
| 1/4 | 500 | 4 | 576.25 | 572.125 | 0.874 | 0.792 | 631.066 | 542.400 | 429.749 |
| 2/1 | 927 | 5 | **(A)** | | 1.589 | 1.627 | 569.586 | 569.068 | 926.156 |
| 2/2 | 500 | 6 | | 594.25 | 0.841 | 0.880 | 568.479 | 595.736 | 523.974 |
| 2/3 | 434 | 7 | 629.5 | 613.875 | 0.707 | **(C)** | 619.422 | 622.404 | 436.089 |
| 2/4 | 532 | 8 | 651.25 | 640.375 | 0.831 | 0.792 | 671.454 | 649.072 | 514.266 |
| 3/1 | 1052 | 9 | 661.75 | 656.5 | 1.602 | 1.627 | 646.392 | 675.740 | 1099.765 |
| 3/2 | 587 | 10 | 667.5 | 664.625 | **(B)** | 0.880 | 667.394 | 702.408 | 617.796 |
| 3/3 | 476 | 11 | 715 | 691.25 | 0.689 | 0.701 | 679.366 | 729.076 | 510.830 |
| 3/4 | 555 | 12 | 729.5 | 722.25 | 0.768 | 0.792 | **(D)** | 755.744 | 598.784 |
| 4/1 | 1242 | 13 | 752.75 | 741.125 | 1.676 | 1.627 | 763.135 | 782.412 | 1273.373 |
| 4/2 | 645 | 14 | 764 | 758.375 | 0.851 | 0.880 | 733.338 | 809.080 | 711.619 |
| 4/3 | 569 | 15 | 827.75 | 795.875 | 0.715 | 0.701 | 812.099 | 835.749 | 585.570 |
| 4/4 | 600 | 16 | 891.5 | 859.625 | 0.698 | 0.792 | 757.279 | 862.417 | 683.301 |
| 5/1 | 1497 | 17 | 927.25 | 909.375 | 1.646 | 1.627 | 919.818 | 889.085 | 1446.982 |
| 5/2 | 900 | 18 | 977.25 | 952.25 | 0.945 | 0.880 | 1023.262 | 915.753 | 805.441 |
| 5/3 | 712 | 19 | | | | 0.701 | 1016.194 | 942.421 | 660.310 |
| 5/4 | 800 | 20 | | | | 0.792 | 1009.706 | 969.089 | 767.819 |
| | | | | **Intercept for 'Deseasonalized Value'** | | | **435.727** | | |
| | | | | **Slope for 'Deseasonalized Value'** | | | **26.668** | | |

**Q9b(i).** Calculate the Moving Average labelled as **(A)** in Table 3b.

**Q9b(ii).** Calculate the Seasonal Ratio labelled as **(B)** in Table 3b.

**Q9b(iii).** State the Seasonal Index labelled as **(C)** in Table 3b.

0.701; 0.70

**Q9b(iv).** Calculate the Deseasonalized Value labelled as **(D)** in Table 3b.

**Q9b(v).** Write down the equation of the regression line used to calculate the Deseasonalized Forecast values.

**Q9b(vi).** Calculate the Seasonalized Forecast for **Month 6 Week 2**.

**Q9b(vii).** Calculate the Absolute Percent Error for the forecast in **Month 4 Week 4**.

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**Q10.** *Show your workings clearly*. ***Round off your answers to 2 decimal places where necessary.***

In a large distribution centre, packages go through labelling before they are shipped. The labelling process is currently done manually by **8 workers**. A recent study found that packages arrive at the labelling station according to a Poisson process at an average rate of 750 packages per hour. Each package takes an average of 30 seconds to label, and the manual labelling time follows a negative exponential distribution.

**Q10a.** Calculate the arrival rate of the packages (in per minute) at the labelling station.

Arrival rate = 750/60 = 12.5

**Q10b.** Calculate the minimum number of workers required at the labelling station for the queuing system to be in a steady state condition.

12.5 < K(2)
12.5 < 2K
6.25 < K
K = 7

**Q10c.** Given that 8 (value of k) workers are present and n is the number of products in the system in Table 4, calculate the probability that **at least 5 of the workers are busy**.

Table 4: Partial Workings of M/M/8 Queuing Model

| n | Probability of n packages in queuing system Pn | |
|---|---|---|
| | $\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$ | $\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$ |
| 0 | 0.002 | 0.747 |
| 1 | 0.011 | 0.583 |
| 2 | 0.035 | 0.456 |
| 3 | 0.073 | 0.356 |
| 4 | 0.114 | 0.278 |
| 5 | 0.143 | 0.217 |
| 6 | 0.149 | 0.170 |
| 7 | 0.133 | 0.133 |
| 8 | 0.104 | 0.104 |
| 9 | 0.072 | 0.081 |
| 10 | 0.045 | 0.063 |

probability = 1 - 0.002 - 0.011 - 0.035 - 0.073 - 0.114

**Q10d.** The labelling supervisor has just purchased an automated labelling machine. The machine performs labelling at a constant time of 10 seconds for every package. If the machine must achieve a utilization of 85%, determine how many packages per hour should be sent to the labelling machine.

**Q10e.** Assuming the arrival of the packages at the labelling machine follows a Poisson process,
i) Identify a suitable queuing model for the queuing system at the labelling machine.
ii) State 2 assumptions of the identified queuing model.

i) M/M/K

**Q10f.** On a particular day, the arrival rate of packages to the labelling machine is 4.8 per minute (labelling time is 10 seconds). Calculate:
i) The average number of packages waiting in the queue, Lq
ii) The average time that the packages wait in the queue, Wq in minutes

**Q11.** Figure 5a shows a simulation snapshot of a goods receiving process modelled in FlexSim software. Incoming items are first unpacked before they are sent to QC. If an item fails QC, it will be sent to Quarantine. If it passes QC, it will go to Stock In.

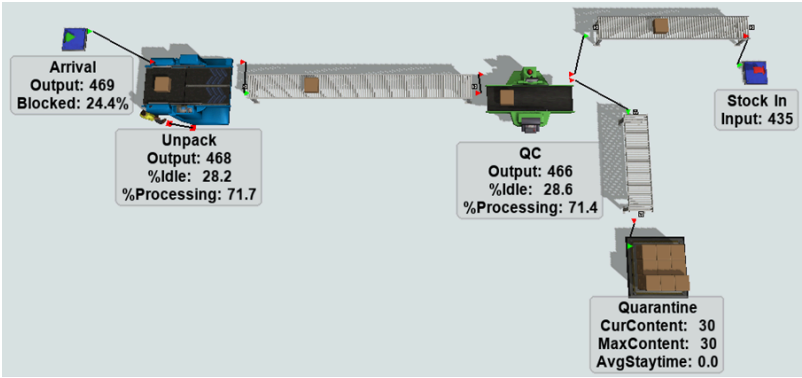


Figure 5a: FlexSim Model of Goods Receiving

**Q11a.** Referring to Figure 5a, state which FlexSim 3D object is used to model the following:

Stock In: sink

QC: processor

**Q11b.** Referring to Figure 5a, state the following statistics:

Number of items in Quarantine = 30

Utilization of Unpack = 71.7; 71.7%; 0.717

**Q11c.** It is known that 6% of incoming items will need to be quarantined. At the QC object, the 'Send to Port' option is set to 'By Percentage' (shown in Figure 5b). Specify the values of A and B in the input box.

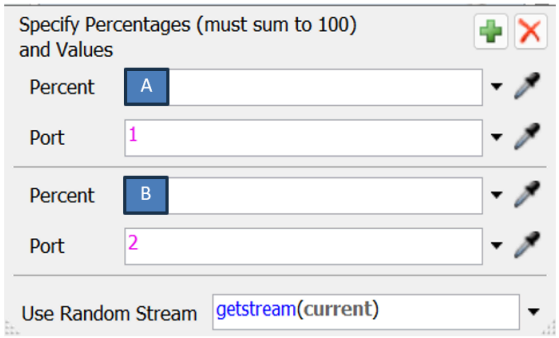


Figure 5b: By Percentage Input Box

A: 96

B: 4

**Q11d.** Figure 5c shows the settings of the 'On Exit' trigger for the QC object. Explain what this trigger does.

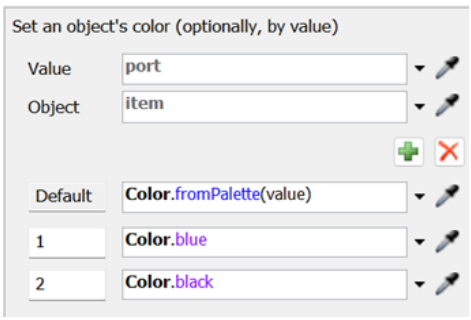


Figure 5c: On Exit Trigger in QC Object

**Q12.** Jack is using a FlexSim Process Flow model to study a customer service process at a call center. Figure 6a shows a snapshot of the simulation.

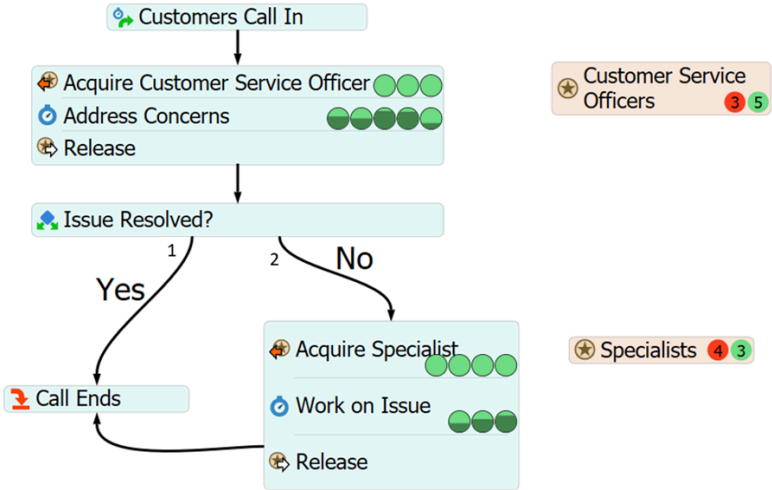


Figure 6a: FlexSim Process Flow Simulation

**Q12a.** How many customers are currently waiting for a Specialist?

4; four

ii) How many Customer Service Officers are there?

5; five

iii) Identify the FlexSim Process Flow activity is used to model 'Issue Resolved?'.

Answer the following questions:i) decide

**Q12b.** Referring to Figure 6b, what values should Jack input for X and Y if 30% of the calls require a specialist to intervene?

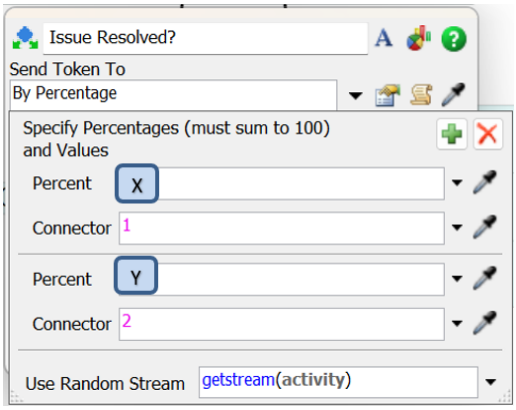


Figure 6b: Activity Properties

X = 70

Y = 30

**Q12c.** Figure 6c shows the statistics for Customer Service Officers and Specialists at the end of a simulation run.

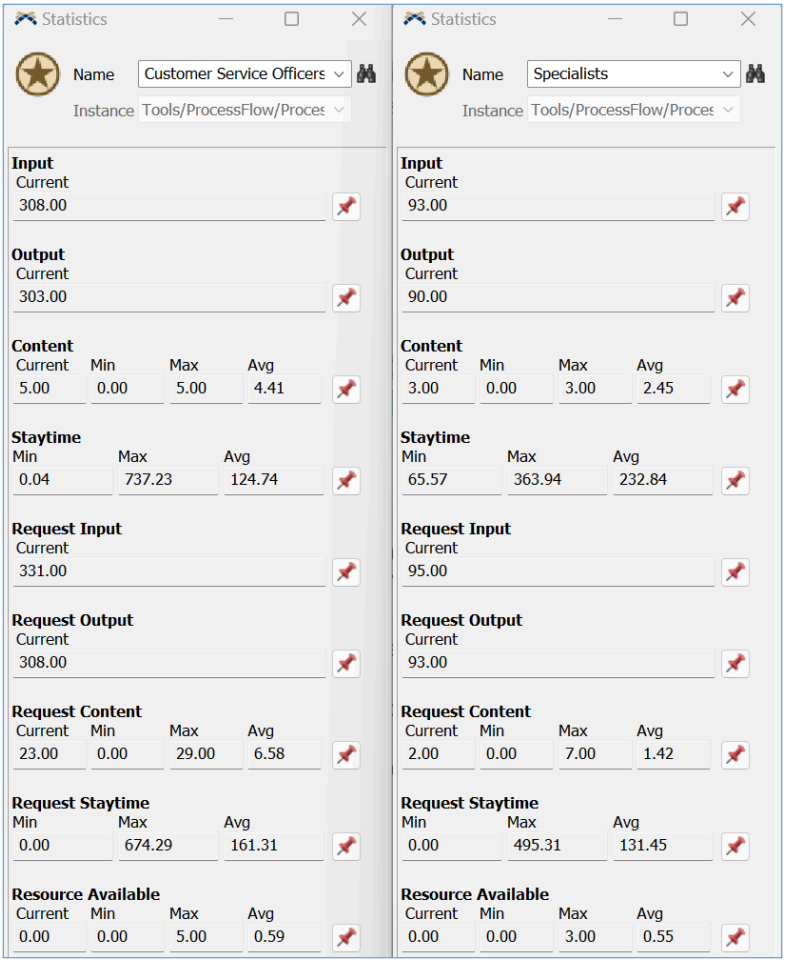


Figure 6c: Statistics of Work Station and QC Station

**Q12c(i).** Determine the bottleneck of the process by comparing the average queue length for the 2 resources.

**Q12c(ii).** Calculate the utilization of the Specialists.

**Q12d.** Someone suggested that Jack should build a 3D model to study the customer service process. Do you agree? Explain why.

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